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MILITARY SPECIFICATION

DIETHYLENE GLYCOL DINITRATE (DEGDN) (FOR USE IN AMMUNITION)

*This specification is approved for use within the U.S. Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1.0 SCOPE

1.1 Scope. - This specification covers the requirements and quality assurance provisions for the manufacture and acceptance of one type of Diethylene Glycol Dinitrate (DEGDN) for use in the formulation of propellants. (See 6.1.)

2.0 APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. - The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

*	MIL-N-246	Nitroglycerin
	MIL-D-23296	Diethylene Glycol
	MIL-A-48078	Ammunition, Standard Quality Assurance Provisions, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC-N/A FSC-1315

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment
MIL-STD-286	Propellant, Solid; Sampling, Inspection and Testing
MIL-STD-1218	ACS Chemicals
MIL-STD-1235	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, drawings, and publications listed above required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

* 2.2 Other publications. - The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is currently on the date of the solicitation.

* AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM E300	Sampling Industrial Chemicals

(Application for copies should be addressed to the American Society For Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

CODE OF FEDERAL REGULATIONS

Title 49	Transportation, Parts 100-199 Special Permit No. 5704

(The Interstate Commerce Commission Regulations are now a part of the Code of Federal Regulations, available from the Superintendent of Documents, U.S. Government Printing Office, Washington D.C. 20402. Orders for the above publications should cite, "49 CFR 100-199 (latest revision)").

* 2.3 Order of precedence. - In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of

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this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3.0 REQUIREMENTS

3.1 Manufacture. - Diethylene Glycol conforming with MIL-D-23296 shall be used in the manufacture of DEGDN.

3.2 Material. - The explosive shall be DEGDN conforming with the following requirements.

* 3.2.1 Nitrogen. - The nitrogen content shall be 14.1 percent minimum to 14.53 percent maximum when determined in accordance with 4.5.1.

* 3.2.2 Moisture. - The moisture content shall be 0.75 percent maximum when determined in accordance with 4.5.2.

* 3.2.3 Alkalinity. - The alkalinity content, calculated as sodium carbonate (Na_2CO_3), shall not exceed .004 percent when determined in accordance with 4.5.3.

* 3.2.4 Acidity. - The acidity content, calculated as sulfuric acid (H_2SO_4), shall not exceed .01 percent when determined in accordance with 4.5.3.

* 3.2.5 Stability. - The DEGDN shall not change the color of standard potassium iodide starch paper in less than ten (10) minutes at 82.2°C when determined in accordance with 4.5.4.

* 3.3 First article. - When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3).

* 3.4 Workmanship. - All parts shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, surface defects, flash, air bubbles, blisters and dirt. The cleaning method shall not be injurious to the parts, nor shall the parts be contaminated by the cleaning agent.

4.0 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. - The inspection requirements specified herein are classified as follows:

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a) First article inspection
b) Quality conformance inspection

4.3 First article inspection.

4.3.1 Submission. - The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of approximately 100 grams, or as otherwise specified by the Contracting Officer of DEGDN obtained by sampling as described in 4.4.3.1. The sample shall be obtained from a production lot which has been produced by the contractor using the same production processes, procedures, equipment and materials which will be used in fulfilling the contract. All materials shall be obtained from the same sources of supply as will be used in regular production.

4.3.2 Inspections to be performed. - The sample will be subjected by the Government (see Table I) to any or all of the examinations or tests specified in 4.5.

*

Table I. First Article Tests.

Test	Requirement	Test Method
Nitrogen Content	3.2.1	4.5.1
Moisture Content	3.2.2	4.5.2
Alkalinity	3.2.3	4.5.3
Acidity	3.2.4	4.5.3
Stability	3.2.5	4.5.4
Workmanship	3.4	Visual

4.3.3 Rejection. - See MIL-A-48078.

4.4 Quality conformance inspection.

4.4.1 Lot formation. - A lot shall consist of one or more batches of DEGDN produced by one manufacturer, in accordance with the same specification, or same specification revision, under one continuous set of operating conditions. Each lot shall consist of that quantity of DEGDN that has been subjected to the same unit chemical or physical mixing process intended to make the final product homogeneous (see MIL-A-48078).

4.4.2 Examination. - The examinations listed in Table II shall be performed on inspection lots as defined in 4.4.1. If the sample fails to comply with any of the requirements, the batch of the lot shall be rejected.

4.4.3 Testing. - PRECAUTION: This specification covers sampling and testing of toxic and hazardous materials. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing the DEGDN. (See 6.3.)

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* 4.4.3.1 Sampling. - The DEGDN shall be sampled in accordance with established procedures of the manufacturing agency. A representative sample of approximately 60 ml shall be removed from each batch of the lot. The DEGDN shall be transferred to a rubber bottle fitted with a stopper (See 6.3) and the sample shall be properly identified. The sample shall be tested in accordance with the applicable methods given in 4.5. If the sample fails to meet any of the test requirements, the batch of the lot shall be rejected. MIL-STD-105 shall apply for defect classification given in Table II.

* 4.4.3.2 Inspection equipment. - For the performance of all tests and examinations specified, commercial inspection equipment should be employed. The contractor shall have available, and utilize correctly, this equipment and is charged with the responsibility of assuring that proper calibration procedures are followed. Government approval of all inspection equipment is required prior to its use for acceptance purposes.

* 4.4.3.3 Tests.

*

Table II. Lot Acceptance Tests.

Test/Examination	Requirement	Defect Classification
Nitrogen	3.2.1	Major
Moisture	3.2.2	Major 1/
Alkalinity	3.2.3	Major
Acidity	3.2.4	Major
Stability	3.2.5	Critical
Workmanship	3.4	Minor

* 1/ When the DEGDN is used at the point of manufacture in the production of propellant by the water slurry method or by the water emulsion method, the moisture requirement shall not apply.

* 4.5. Test methods and procedures. - All tests given in this section shall be performed using prescribed procedures for replicate determinations given in standard analytical chemistry textbooks. Also, unless otherwise specified herein, all chemicals shall be Reagent Grade or ACS Grade in accordance with MIL-STD-1218.

* 4.5.1 Nitrogen. - The nitrogen shall be determined in accordance with Method 209.3 of Standard MIL-STD-286 or equivalent Method T 209.11.

* 4.5.2 Moisture. - The moisture shall be determined in accordance with Method 101.5 of Standard MIL-STD-286.

* 4.5.3 Acidity or alkalinity. - The acidity or alkalinity shall be determined in accordance with Method 107.1 of Standard MIL-STD-286 using tolulene as the solvent.

* 4.5.4 Determination of stability. - The stability shall be determined in accordance with MIL-N-246, paragraph 4.3.4.

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5.0 PACKAGING

5.1 Preservation, packing and packaging. - Diethylene Glycol Dinitrate is classed as a "Forbidden Explosive" in Code of Federal Regulations, Title 49, parts 100-199, for the Transportation of Explosives and Other Articles by Freight. For shipment by public highways see CFR, Title 49, section 173.51 and Special Permit No. 5704 of the Department of Transportation (DOT) Hazardous Materials Regulations.

5.2 Marking. - The marking of interior packages and exterior shipping containers shall be in accordance with instructions contained in Special Permit No. 5704, Interstate Commerce Commission Regulations and in accordance with Standard MIL-STD-129.

6.0 NOTES

* 6.1 Intended use. - The DEGDN of this specification is intended for use as a propellant ingredient.

6.2 Ordering data. - See MIL-A-48078.

6.3 Toxicity. - The DEGDN may cause dilation of blood vessels, headache, nausea, sinuses, and reduce blood pressure. Extreme exposures resulting from inhalation, skin contact or ingestion may result in central nervous system depression, coma, and respiratory paralysis. Drinking of any quantity of the liquids must be avoided. Severe headache shortly after exposure may diminish after development of tolerance. Ventilation should be used to reduce the inhalation hazard. Protective gloves and aprons are helpful in avoiding skin contact. Frequent washing in soap and water will minimize skin absorption. Alcohol or other solvents should not be used on the skin. The threshold limit value ceiling is 0.2 ppm.

* 6.4 Submission of inspection equipment designs for approval. (see MIL-A-48078). - Submit equipment designs, as required to Commander, U.S. Army Armament Research, Development and Engineering Center, ATTN: AMSMC-QAT-I-(D), Dover, New Jersey 07801-5001.

* 6.5 Equivalent test methods. - Prior approval of the Contracting Officer is required for use of an equivalent test method. A description of the proposed method should be submitted through the Contracting Officer to: Commander, ARDEC, ATTN: AMSMC-QAR-(D), Dover, New Jersey 07801-5001. This description should include but not be limited to the procedures used, the accuracy and precision of the method currently used versus the proposed method, test data to demonstrate the accuracy and precision and drawings of any special equipment required.

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* 6.6 Contract data requirements. - One copy of all description sheets and firing records required by this specification shall be forwarded to:

Commander
U.S. Army Armament Research, Development and Engineering Center
U.S. Army Armament Munitions and Chemical Command
ATTN: AMSMC-QAR-R(D)
AMSMC-LCA-G(D)
Dover, NJ 07801-5001

Commander
U.S. Army Armament Munitions and Chemical Command
ATTN: AMSMC-QAD(R)
Rock Island, IL 61299.

* 6.7 Keywords. -
Ammunition
Cartridge
Cannon
Diethylene Glycol Dinitrate (DEGDN)
Tank

* 6.8 The margins of this standard are marked with asterisks to indicate where significant changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations.

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